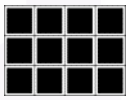
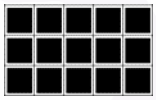
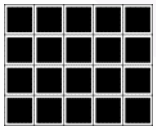
General Aptitude (GA)

Q.1 – Q.5 Carry ONE mark Each

Q.1	The antonym of the word protagonist is _____.
(A)	agnostic
(B)	antagonist
(C)	arsonist
(D)	anarchist

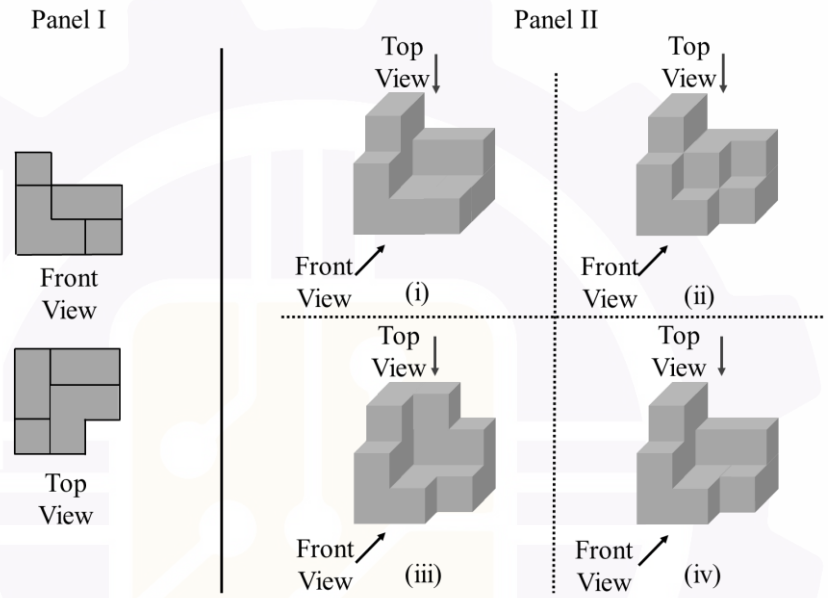
Q.2	The figure shows two 4-tile patterns.  Either one or both of the patterns can be used any number of times and in any orientation to construct a new pattern. Which one of the options below **cannot** be constructed by using only these two 4-tile patterns assuming there are no overlaps among them?
(A)	
(B)	
(C)	
(D)	

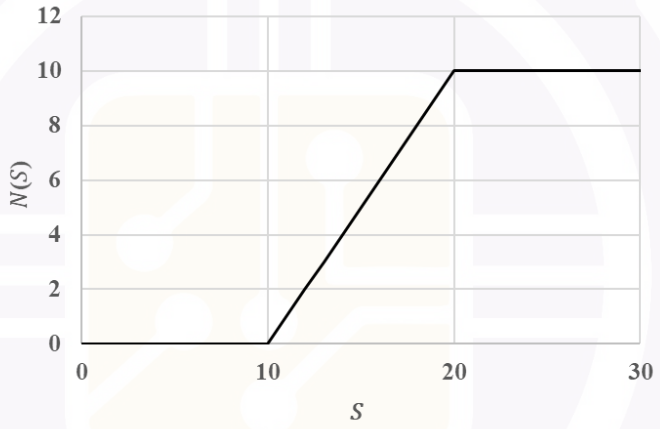
Q.3	Consider a knock-out women's badminton singles tournament where there are no ties. The loser in each game is eliminated from the tournament. Every player plays until she is defeated or remains the last undefeated player. The last undefeated player is declared the winner of the tournament. If there are 64 players in the beginning of the tournament, how many games should be played in total to declare the winner of the tournament?
(A)	127
(B)	64
(C)	63
(D)	32

Q.4	A student needs to enroll for a minimum of 60 credits. A student cannot enroll for more than 70 credits. The credits are divided amongst project and three distinct sets of courses namely, core courses, specialization courses, and elective courses. It is compulsory for a student to enroll for exactly 15 credits of core courses and exactly 20 credits of project. In addition, a student has to enroll for a minimum of 10 credits of specialization courses. The maximum credits of elective courses that a student can enroll for is _____
(A)	10
(B)	15
(C)	20
(D)	25
Q.5	‘When the teacher is in the room, all students stand silently.’ If the above statement is true, which one of the following statements is **not necessarily** true?
(A)	If any student is not standing silently, then the teacher is not in the room.
(B)	When the teacher is in the room, all students are silent.
(C)	If all students are standing, then the teacher is in the room.
(D)	When the teacher is in the room, all students are standing.

Q.6 – Q.10 Carry TWO marks Each

Q.6	Combinatorics deals with problems involving counting. For example, “How many distinct arrangements of N distinct objects in M spaces on a circle are possible?” is a typical problem in combinatorics. This kind of counting is sometimes used in the modeling of several physical phenomena. Often, in such models, the different combinatorial possibilities are assigned probability values. Assigning probabilities enables the computation of the average values of physical quantities. Consider the following statements: P: Combinatorics is always invoked in the modeling of physical phenomena. Q: Modeling some physical phenomena involves assigning probabilities to combinatorial possibilities in order to compute average values of physical quantities. Based on the passage above, what can be inferred about statements P and Q?
(A)	P is False and Q is False
(B)	P is False and Q is True
(C)	P is True and Q is False
(D)	P is True and Q is True

Q.7	In Panel I of the figure below, the front view and top view of a structure are shown. Which one of the 3D structures shown in Panel II possesses the views shown in Panel I? 
(A)	(i)
(B)	(ii)
(C)	(iii)
(D)	(iv)

Q.8	For positive real numbers S and K, the function $H_K(S)$ is defined as: $H_K(S) = \max(S - K, 0)$. The max function is defined as: $$\max(a, b) = \begin{cases} a, & \text{when } a > b \\ b, & \text{when } a \leq b \end{cases}$$ The graph below shows the plot of a function $N(S)$ versus S. $N(S)$ can be expressed as _____. 
(A)	$H_{10}(S) - H_{20}(S)$
(B)	$H_{10}(S) - 2H_{20}(S)$
(C)	$-H_{10}(S) + H_{20}(S)$
(D)	$H_{15}(S) - H_{20}(S)$

Q.9	In the 2020 summer Olympics' Javelin throw finals, Neeraj Chopra exhibited a spectacular performance to win the gold medal. The silver medal was won by Jakub Vadlejch and the bronze medal was won by Vitezlav Vesely. There were six rounds of throws with each athlete having one throw per round. The best of all the throws of each athlete is considered for the medal. Following were the observations about the throws: i. The first and second rounds were dominated by Neeraj Chopra with a gold medal performance in his second throw, while the other two athletes did not have any medal winning throws in these rounds. ii. The throws in the last round by both Jakub Vadlejch and Vitezlav Vesely were fouls and were not considered for scoring. iii. After four rounds, Vitezlav Vesely was in the second position and could not improve upon his best throw in the succeeding rounds. iv. In the fourth round, the throw by Jakub Vadlejch was the best in that round. In which round did Vitezlav Vesely have his best throw?
(A)	Third
(B)	Fourth
(C)	Fifth
(D)	Sixth

Q.10	An unbiased six-faced dice whose faces are marked with numbers 1, 2, 3, 4, 5, and 6 is rolled twice in succession and the number on the top face is recorded each time. The probability that the number appearing in the second roll is an integer multiple of the number appearing in the first roll is _____
(A)	$\frac{1}{6}$
(B)	$\frac{5}{18}$
(C)	$\frac{7}{18}$
(D)	$\frac{5}{6}$

Q.11 – Q.35 Carry ONE mark Each

Q.11	An urn contains one red ball and one blue ball. At each step, a ball is picked uniformly at random from the urn, and this ball together with another ball of the same color is put back in the urn. The probability that there are equal number of red and blue balls after two steps is
(A)	1/4
(B)	1/3
(C)	1/2
(D)	2/3
Q.12	Consider 4×4 matrices with their elements from $\{\mathbf{0}, \mathbf{1}\}$. The number of such matrices with even number of **1**s in every row and every column is
(A)	512
(B)	1025
(C)	1023
(D)	255

Q.13	For $n > 1$, the maximum multiplicity of any eigenvalue of an $n \times n$ matrix with elements from $\mathbb{R}$ is
(A)	n
(B)	$n - 1$
(C)	1
(D)	$n + 1$
Q.14	Match each addressing mode in **List I** with a data element or an element of a data structure (in a high-level language) in **List II**:

List I	List II
P. Immediate	1. Element of an array
Q. Indirect	2. Pointer
R. Base with index	3. Element of a record
S. Base with offset/displacement	4. Constant

(A)	P–4, Q–3, R–1, S–2
(B)	P–4, Q–2, R–1, S–3
(C)	P–1, Q–4, R–3, S–2
(D)	P–2, Q–3, R–1, S–4

Q.15	Consider a processor P whose instruction set architecture is the load-store architecture. The instruction format is such that the first operand of any instruction is the destination operand. Which one of the following sequences of instructions corresponds to the high-level language statement Z = X + Y ? *Note*: X, Y, and Z are memory operands. R0, R1, and R2 are registers.
(A)	ADD Z, X, Y
(B)	LOAD R0, X ADD Z, R0, Y
(C)	ADD R0, X, Y STORE Z, R0
(D)	LOAD R0, X LOAD R1, Y ADD R2, R0, R1 STORE Z, R2

Q.16	Which one of the following dependencies among the register operands of different instructions can cause a data hazard in a pipelined processor?
(A)	Read-after-read
(B)	Read-after-write
(C)	Write-after-read
(D)	Write-after-write

Q.17	Consider the following recurrence relations: For all $n > 1$, $$T_1(n) = 4T_1\left(\frac{n}{2}\right) + T_2(n)$$ $$T_2(n) = 5T_2\left(\frac{n}{4}\right) + \Theta(\log_2 n)$$ Assume that for all $n \leq 1, T_1(n) = 1$ and $T_2(n) = 1$. Which one of the following options is correct?
(A)	$T_1(n) = \Theta(n^2)$
(B)	$T_1(n) = \Theta(n^2 \log_2 n)$
(C)	$T_1(n) = \Theta(n^{\log_4 5})$
(D)	$T_1(n) = \Theta(n^{\log_4 5} \log_2 n)$

Q.18	With respect to a TCP connection between a client and a server, which one of the following statements is true?
(A)	The client and server use a two-way handshake mechanism before the start of data transmission
(B)	The server cannot initiate closing of the connection before the client initiates closing of the connection
(C)	The TCP connection is half-duplex
(D)	The client and server can initiate closing of the connection at the same time
Q.19	Which of the following statements is/are true with respect to the interaction of a web browser with a web server using HTTP 1.1?
(A)	HTTP 1.1 facilitates downloading multiple objects of the same webpage over the same TCP connection, if the objects are stored in the same server
(B)	HTTP 1.1 facilitates downloading multiple objects of the same webpage over the same TCP connection, even if they are stored in different servers
(C)	HTTP 1.1 facilitates sending a request for downloading one object without waiting for a previously requested object to be downloaded completely
(D)	HTTP 1.1 facilitates downloading multiple webpages on the same server to be downloaded over a single TCP connection

Q.20	Let $n > 1$. Consider an $n \times n$ matrix M with its elements from $\mathbb{R}$. Let the vector $(0, 1, 0, 0, \ldots, 0) \in \mathbb{R}^n$ be in the null space of M. Which of the following options is/are always correct?
(A)	Determinant of M is 1
(B)	Determinant of M is 0
(C)	Rank of M is 1
(D)	There are at least two non-zero vectors in the null space of M
Q.21	Consider the following Boolean expression of a function F : $$F(P, Q) = (\bar{P} + Q) \oplus (\bar{P}Q)$$ Which of the following expressions is/are equivalent to F ?
(A)	$\overline{P \oplus Q}$
(B)	$P \oplus Q$
(C)	$\bar{P} \oplus Q$
(D)	$\bar{P} \oplus \bar{Q}$

Q.22	Consider the 8-bit signed integers X, Y and Z represented using the sign-magnitude form. The binary representations of X and Y are as follows: X: 10110100 Y: 01001100 Which of the following operations to compute Z result(s) in an arithmetic overflow?
(A)	$Z = X + Y$
(B)	$Z = X - Y$
(C)	$Z = -X + Y$
(D)	$Z = -X - Y$

Q.23	Let n be an odd number greater than 100. Consider a binary minheap with n elements stored in an array P whose index starts from 1. Which of the following indices of P do/does NOT correspond to any leaf node of the minheap?
(A)	$\frac{n+1}{2}$
(B)	$\frac{n-1}{2}$
(C)	$\frac{n-3}{2}$
(D)	n

Q.24	Consider a hash table $P[0, 1, \ldots, 10]$ that is initially empty. The hash table is maintained using open addressing with linear probing. The hash function used is $h(x) = (x + 7) \bmod 11$. Consider the following sequence of insertions performed on P: $1, 13, 22, 15, 11, 24$ Which of the following positions in the hash table is/are empty after these insertions are performed?
(A)	0
(B)	10
(C)	2
(D)	1

Q.25	Consider the following grammar where S is the start symbol, and a and b are terminal symbols. $$S \rightarrow aSbS \mid bS \mid \epsilon$$ Which of the following statements is/are true?
(A)	The grammar is ambiguous
(B)	The string abb has two distinct derivations in this grammar
(C)	The string $abab$ has only one rightmost derivation
(D)	The language generated by the grammar is undecidable
Q.26	Let M be a nondeterministic finite automaton (NFA) with 6 states over a finite alphabet. Which of the following options CANNOT be the number of states in the minimal deterministic finite automaton (DFA) that is equivalent to M ?
(A)	32
(B)	65
(C)	1
(D)	128

Q.27	Consider the following C statements:
	`char *str1 = "Hello;  /* Statement S1 */`
	`char *str2 = "Hello;";  /* Statement S2 */`
	`int *str3 = "Hello";  /* Statement S3 */`
	Which of the following options is/are correct?

(A)	`S1` and `S2` have syntactic errors
(B)	`S2` has a lexical error and `S3` has a syntactic error
(C)	`S1` has a lexical error and `S3` has a semantic error
(D)	`S1` has a syntactic error and `S3` has a semantic error
Q.28	Which of the following statements is/are true?
(A)	LL(1) parser uses backtracking
(B)	For a grammar to be LL(1), it must be left-recursive
(C)	For a grammar to be LL(1), it must be left-factored
(D)	The LL(1) parsers are more powerful than the SLR parsers

Q.29	With respect to deadlocks in an operating system, which of the following statements is/are FALSE?
(A)	Banker's algorithm is used to prevent deadlocks
(B)	Deadlock formation can be prevented by ensuring that the hold and wait condition is not allowed
(C)	An assignment edge in a resource allocation graph is marked from a process to a resource
(D)	A safe state guarantees that all processes can finish without formation of a deadlock
Q.30	Let P, Q, R and S be the attributes of a relation in a relational schema. Let $X \longrightarrow Y$ indicate functional dependency in the context of a relational database, where $X, Y \subseteq \{P, Q, R, S\}$. Which of the following options is/are always true?
(A)	If $\big(\{P, Q\} \longrightarrow \{R\}$ and $\{P\} \longrightarrow \{R\}\big)$, then $\{Q\} \longrightarrow \{R\}$
(B)	If $\{P, Q\} \longrightarrow \{R\}$, then $\big(\{P\} \longrightarrow \{R\}$ or $\{Q\} \longrightarrow \{R\}\big)$
(C)	If $\big(\{P\} \longrightarrow \{R\}$ and $\{Q\} \longrightarrow \{S\}\big)$, then $\{P, Q\} \longrightarrow \{R, S\}$
(D)	If $\{P\} \longrightarrow \{R\}$, then $\{P, Q\} \longrightarrow \{R\}$

Q.31	In the context of relational database normalization, which of the following statements is/are true?
(A)	It is always possible to obtain a dependency-preserving 3NF decomposition of a relation
(B)	It is always possible to obtain a dependency-preserving 1NF decomposition of a relation
(C)	It is not always possible to obtain a dependency-preserving BCNF decomposition of a relation
(D)	It is not always possible to obtain a dependency-preserving 2NF decomposition of a relation
Q.32	Consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined as follows: $$f(x) = \begin{cases} c_1 e^x - c_2 \log_e\left(\frac{1}{x}\right), & \text{if } x > 0 \\ 3 & \text{otherwise} \end{cases}$$ where $c_1, c_2 \in \mathbb{R}$. If f is continuous at $x = 0$, then $c_1 + c_2$ = _____. (*answer in integer*)
Q.33	The height of a binary tree is the number of edges in the longest path from the root to a leaf in the tree. The maximum possible height of a full binary tree with 23 nodes is _____. (*answer in integer*)

Q.34	Consider the following program in C:
	
	`#include <stdio.h>`
	
	`void func(int i, int j) {`
	`  if(i < j) {`
	`  int i = 0;`
	`  while (i < 10) {`
	`  j += 2;`
	`  i++;`
	`  }`
	`  }`
	`  printf("%d", i);`
	`}`
	
	`int main() {`
	`  int i = 9, j = 10;`
	`  func(i, j);`
	`  return 0;`
	`}`
	
	The output of the program is _____. (*answer in integer*)
	
	Note: Assume that the program compiles and runs successfully.

Q.35	Consider a system consisting of k instances of a resource R, being shared by 5 processes. Assume that each process requires a maximum of two instances of resource R and a process can request or release only one instance at a time. Further, a process can request the second instance of the resource only after acquiring the first instance. The minimum value of k for the system to be deadlock-free is _____. (*answer in integer*)

Q.36 – Q.65 Carry TWO marks Each

Q.36	Consider the real valued variables X, Y and Z represented using the IEEE 754 single-precision floating-point format. The binary representations of X and Y in hexadecimal notation are as follows: X: 35C00000 Y: 34A00000 Let $Z = X + Y$. Which one of the following is the binary representation of Z, in hexadecimal notation?
(A)	35C80000
(B)	35CC0000
(C)	35E80000
(D)	35EC0000

Q.37 Consider a 2-bit saturating up/down counter that performs the saturating up count when the input P is 0, and the saturating down count when P is 1. The Next State table of the counter is as shown. The counter is built as a synchronous sequential circuit using D flip-flops.

Input	Current State		Next State	
P	Q_1	Q_0	Q_1^+	Q_0^+
0	0	0	0	1
0	0	1	1	0
0	1	0	1	1
0	1	1	1	1
1	0	0	0	0
1	0	1	0	0
1	1	0	0	1
1	1	1	1	0

Which one of the following options corresponds to the expressions for the inputs of the D flip-flops, D_1 and D_0?

(A) $D_1 = P\,Q_1 + \bar{P}Q_0 + Q_1Q_0$ $\quad$ $D_0 = P\,Q_0 + \bar{P}\,Q_1 + Q_1\,\overline{Q_0}$

(B) $D_1 = \bar{P}\,Q_1 + \bar{P}Q_0 + Q_1Q_0$ $\quad$ $D_0 = \bar{P}\,\overline{Q_0} + \bar{P}\,Q_1 + Q_1\,\overline{Q_0}$

(C) $D_1 = \bar{P}\,\overline{Q_1} + \bar{P}\,Q_0 + Q_1\,Q_0$ $\quad$ $D_0 = \bar{P}\,Q_0 + \bar{P}\,Q_1 + Q_1\,\overline{Q_0}$

(D) $D_1 = P\,\overline{Q_1} + \bar{P}\,Q_0 + Q_1\,Q_0$ $\quad$ $D_0 = P\,\overline{Q_0} + \bar{P}\,Q_1 + Q_1\,\overline{Q_0}$

Q.38	The size of the physical address space of a processor is 2^{32} bytes. The capacity of a cache memory unit is 2^{23} bytes. The cache block size is 128 bytes. The cache memory unit can be built as a direct mapped cache or as a K-way set-associative cache, where $K = 2^L$ and $L \in \{1, 2, 3\}$. Let the length of the TAG field be M bits for the direct mapped cache, and N bits for the set-associative cache. Which one of the following options is true?
(A)	$N = M + L$
(B)	$N = M - L$
(C)	$N = M + K$
(D)	$N = M - K$

Q.39

Consider the following code snippet in C language that computes the number of nodes in a non-empty singly linked list pointed to by the pointer variable `head`.

```
struct node{
  int elt;
  struct node *next;
};

int getListSize (struct node *head)
{
  if( E1 ) return 1;
  return E2;
}
```

Which one of the following options gives the correct replacements for the expressions **E1** and **E2**?

(A) **E1**: `head == NULL`
E2: `1 + getListSize(head)`

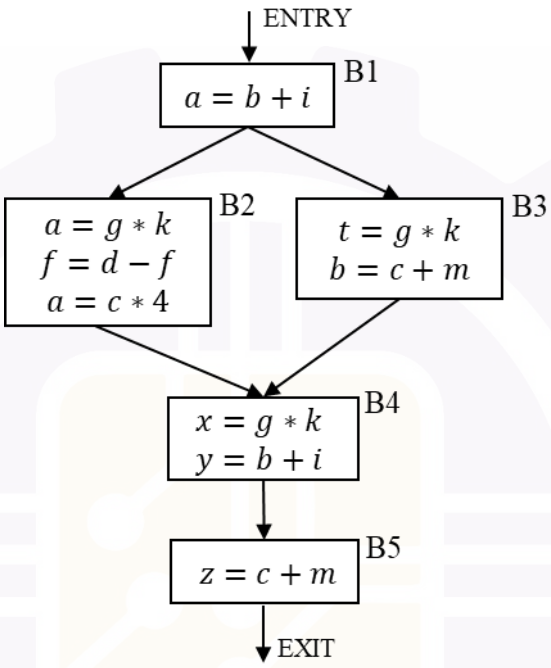
(B) **E1**: `head->next == NULL`
E2: `1 + getListSize(head->next)`

(C) **E1**: `head == NULL`
E2: `1 + getListSize(head->next)`

(D) **E1**: `head->next == NULL`
E2: `1 + getListSize(head)`

Q.40	Let P be the set of all integers from 1 to 15. Consider any order of insertion of the elements of P into a binary search tree that creates a complete binary tree. Which one of the following elements can NEVER be the third element that is inserted?
(A)	4
(B)	2
(C)	10
(D)	5

Q.41	Let $G(V, E)$ be an undirected, edge-weighted graph with integer weights. The weight of a path is the sum of the weights of the edges in that path. The length of a path is the number of edges in that path. Let $s \in V$ be a vertex in G. For every $u \in V$ and for every $k \geq 0$, let $d_k(u)$ denote the weight of a shortest path (in terms of weight) from s to u of length at most k. If there is no path from s to u of length at most k, then $d_k(u) = \infty$. Consider the statements: S1: For every $k \geq 0$ and $u \in V$, $d_{k+1}(u) \leq d_k(u)$. S2: For every $(u, v) \in E$, if (u, v) is part of a shortest path (in terms of weight) from s to v, then for every $k \geq 0, d_k(u) \leq d_k(v)$. Which one of the following options is correct?
(A)	Only S1 is true
(B)	Only S2 is true
(C)	Both S1 and S2 are true
(D)	Neither S1 nor S2 is true

Q.42	Consider the control flow graph shown in the figure.  Which one of the following options correctly lists the set of redundant expressions (common subexpressions) in the basic blocks B4 and B5? *Note:* All the variables are integers.
(A)	B4: { $b + i$ } B5: { $c + m$ }
(B)	B4: { $g * k$ } B5: { $c + m$ }
(C)	B4: { $g * k,\ b + i$ } B5: { }
(D)	B4: { $g * k$ } B5: { }

Q.43	Consider a relational database schema with two relations $R(P, Q)$ and $S(X, Y)$. Let $E = \{\langle u \rangle \mid \exists v \, \exists w \; \langle u, v \rangle \in R \wedge \langle v, w \rangle \in S\}$ be a tuple relational calculus expression. Which one of the following relational algebraic expressions is equivalent to E ?
(A)	$\Pi_P(R \bowtie_{R.P=S.X} S)$
(B)	$\Pi_P\left(S \bowtie_{S.X=R.Q} R\right)$
(C)	$\Pi_P(R \bowtie_{R.P=S.Y} S)$
(D)	$\Pi_P\left(S \bowtie_{S.Y=R.Q} R\right)$

Q.44	A TCP sender successfully establishes a connection with a TCP receiver and starts the transmission of segments. The TCP congestion control mechanism's slow-start threshold is set to 10000 segments. Assume that the round-trip time is fixed at 1 millisecond. Assume that the sender always has data to send, the segments are numbered from 1, and no segment is lost. Let t denote the time (in milliseconds) at which the transmission of segment number 2000 starts. Which one of the following options is correct?
(A)	$9 \leq t < 10$
(B)	$10 \leq t < 11$
(C)	$11 \leq t < 12$
(D)	$12 \leq t < 13$

Q.45	Consider the implementation of sliding window protocol over a lossless link, with a window size of W frames, where each frame is of size 1000 bits (including header). The bandwidth of the link is 100 kbps ($1k = 10^3$) and the one-way propagation delay is 100 milliseconds. Assume that processing times at the sender and receiver are zero and the transmission time of acknowledgements is also zero. Which one of the following options gives the minimum size of W (in number of frames) required to achieve 100% link utilization?
(A)	10
(B)	21
(C)	20
(D)	11

Q.46	Let $f: \mathbb{R} \to \mathbb{R}$ be defined as follows: $$f(x) = \left(\frac{\lvert x \rvert}{2} - x\right)\left(x - \frac{\lvert x \rvert}{2}\right)$$ Which of the following statements is/are true?
(A)	f has a local maximum
(B)	f has a local minimum
(C)	f' is continuous over $\mathbb{R}$
(D)	f' is not differentiable over $\mathbb{R}$
Q.47	Let $G(V, E)$ be a simple, undirected graph. A vertex cover of G is a subset $V' \subseteq V$ such that for every $(u, v) \in E$, $u \in V'$or $v \in V'$. Let the size of the smallest vertex cover in G be k. Let S be any vertex cover of size k. For a vertex $v \in V$, which of the following constraints will always ensure that $v \in S$?
(A)	The degree of v is at least $k + 1$
(B)	The vertex v is on a path of length $k + 1$
(C)	The vertex v is on a cycle of length $k + 1$
(D)	The vertex v is a part of a clique of size k

Q.48	Consider a Boolean function F with the following minterm expression: $$F(P,Q,R,S) = \sum m\,(1,2,3,4,5,7,10,12,13,14)$$ Which of the following options is/are the minimal sum-of-products expression(s) of F ?
(A)	$\bar{P}S + Q\bar{R} + \bar{P}\bar{Q}R + \bar{Q}R\bar{S}$
(B)	$\bar{P}S + Q\bar{R} + \bar{P}\bar{Q}R + PR\bar{S}$
(C)	$\bar{P}S + Q\bar{R} + PQ\bar{S} + PR\bar{S}$
(D)	$\bar{P}S + Q\bar{R} + PQ\bar{S} + \bar{Q}R\bar{S}$

Q.49	Let $G(V, E)$ be a simple, undirected, edge-weighted graph with unique edge weights. Which of the following statements about the minimum spanning trees (MST) of G is/are true?
(A)	In every cycle C of G, the edge with the largest weight in C is not in any MST
(B)	In every cycle C of G, the edge with the smallest weight in C is in every MST
(C)	For every vertex $v \in V$, the edge with the largest weight incident on v is not in any MST
(D)	For every vertex $v \in V$, the edge with the smallest weight incident on v is in every MST

Q.50 Consider the following pseudocode for depth-first search (DFS) algorithm which takes a directed graph $G(V, E)$ as input, where $d[v]$ and $f[v]$ are the discovery time and finishing time, respectively, of the vertex $v \in V$.

```
DFS(G):
  unmark all v ∈ V
  t ← 0
  for each v ∈ V
     if v is unmarked
        t ← Explore(G, v, t)
     end if
  end for
```

```
Explore(G, v, t):
   mark v
     t ← t + 1
     d[v] ← t
     for each (v, w) ∈ E
          if w is unmarked
               t ← Explore(G, w, t)
          end if
     end for
     t ← t + 1
     f[v] ← t
     return t
```

Suppose that the input directed graph $G(V, E)$ is a directed acyclic graph (DAG).

For an edge $(u, v) \in E$, which of the following options will NEVER be correct?

(A) $d[u] < d[v] < f[v] < f[u]$

(B) $d[v] < d[u] < f[u] < f[v]$

(C) $d[v] < f[v] < d[u] < f[u]$

(D) $d[u] < d[v] < f[u] < f[v]$

Q.51	Let L_1 and L_2 be two languages over a finite alphabet, such that $L_1 \cap L_2$ and L_2 are regular languages. Which of the following statements is/are always true?
(A)	L_1 is regular
(B)	$L_1 \cup L_2$ is regular
(C)	$\overline{L_2}$ is context-free
(D)	L_1 is context-free

Q.52	Consider the following context-free grammar G. $S \rightarrow abaABAbba$ $A \rightarrow aaBBAb \mid bBabaa$ $B \rightarrow aBb \mid ab$ In the above grammar, S is the start symbol, a and b are terminal symbols, and A and B are non-terminal symbols. Let $L(G)$ be the language generated by the grammar G. For a string $s \in L(G)$, let $n_1(s)$ be the number of a's in s and $n_2(s)$ be the number of b's in s. Which of the following statements is/are true?
(A)	There is a string $s \in L(G)$ such that $n_1(s) < n_2(s)$
(B)	For every string $s \in L(G)$, $n_1(s) \geq n_2(s)$
(C)	There is a string $s \in L(G)$ such that $n_1(s) > 2n_2(s)$
(D)	For every string $s \in L(G)$, $n_1(s) \leq 2n_2(s)$

Q.53 Consider the following two syntax-directed definitions SDD1 and SDD2 for type declarations.

SDD1	
Grammar (G1)	Semantic Rules
$D \rightarrow T\ V$	$D.type = T.type$ $V.type = T.type$
$T \rightarrow int$	$T.type = int$
$T \rightarrow float$	$T.type = float$
$V \rightarrow V_1\ id$	$V_1.type = V.type$ $put(id.entry, V.type)$
$V \rightarrow id$	$put(id.entry, V.type)$

SDD2	
Grammar (G2)	Semantic Rules
$D \rightarrow D_1\ id$	$D.type = D_1.type$ $put(id.entry, D_1.type)$
$D \rightarrow T\ id$	$D.type = T.type$ $put(id.entry, T.type)$
$T \rightarrow int$	$T.type = int$
$T \rightarrow float$	$T.type = float$

D is the start symbol, and int, $float$ and id are the three terminals. The non-terminal V_1 is the same as V and the non-terminal D_1 is the same as D. Here, the subscript is used to differentiate the grammar symbols on the two sides of a production. The function put updates the symbol table with the type information for an identifier.

Let P and Q be the languages specified by grammars G1 and G2, respectively.

Which of the following statements is/are true?

(A) The languages P and Q are the same

(B) SDD2 is S-attributed and contains only synthesized attributes

(C) SDD1 is L-attributed and contains only inherited attributes

(D) The specifications of SDD1 and SDD2 are such that the same entries get added to the symbol table

Q.54	Consider a system that has a cache memory unit and a memory management unit (MMU). The address input to the cache memory is a physical address. The MMU has a translation lookaside buffer (TLB). Assume that when a page is evicted from the main memory, the corresponding blocks in the cache are marked as invalid. For a given memory reference, which of the following sequences of events can NEVER happen?
(A)	TLB miss, Page table hit, Cache hit
(B)	TLB hit, Page table miss, Cache hit
(C)	TLB miss, Page table miss, Cache hit
(D)	TLB miss, Page table miss, Cache miss

Q.55	An undirected, unweighted, simple graph $G(V, E)$ is said to be 2-colorable if there exists a function $c: V \rightarrow \{0, 1\}$ such that for every $(u, v) \in E$, $c(u) \neq c(v)$. Which of the following statements about 2-colorable graphs is/are true?
(A)	If G is 2-colorable, then G may contain cycles of odd length
(B)	If G is 2-colorable, then G may contain cycles of even length
(C)	An optimal algorithm for testing whether G is 2-colorable runs in time $\Theta(\|V\| + \|E\|)$, if G is represented as an adjacency list
(D)	An optimal algorithm for testing whether G is 2-colorable runs in time $\Theta(\|E\| \log \|V\|)$, if G is represented as an adjacency list
Q.56	An ISP having an address block 202.16.0.0/15 assigns a block of 6000 IP addresses to a client, using the classless internet domain routing (CIDR) super-netting approach. Which of the following address blocks can be assigned by the ISP?
(A)	202.16.0.0/19
(B)	202.17.64.0/19
(C)	202.16.32.0/19
(D)	202.17.24.0/19

Q.57 Let G be an undirected graph, which is a path on 8 vertices. The number of matchings in G is _____. (*answer in integer*)

Q.58 Let X be a random variable which takes values in the set $\{1, 2, 3, 4, 5, 6, 7, 8\}$. Further, $\Pr(X = 1) = \Pr(X = 2) = \Pr(X = 5) = \Pr(X = 7) = \frac{1}{6}$ and $\Pr(X = 3) = \Pr(X = 4) = \Pr(X = 6) = \Pr(X = 8) = \frac{1}{12}$.

The expected value of X, denoted by $E[X]$, is equal to _____. (*rounded off to two decimal places*)

Q.59 Consider a hard disk with a rotational speed of 15000 rpm. The time to move the read/write head from a track to its adjacent track is 1 millisecond. Initially, the head is on track 0. The number of sectors per track is 400. The sector size is 1024 bytes. It is necessary to transfer data from 10 randomly located sectors in each of the following tracks in the order: 5, 12 and 7.

The total time for the data transfer (in milliseconds) from the hard disk is _____. (*rounded off to one decimal place*)

Q.60 The EX stage of a pipelined processor performs the memory read operations for LOAD instructions, and the operations for the arithmetic and logic instructions. Let t_{EX} denote the time taken by the EX stage to perform the operation for an instruction. For each instruction type, the values of t_{EX} and M (the number of instructions of that type in a sequence of 100 instructions for a program P), are given in the table below.

The duration of the pipeline clock cycle is 1 nanosecond. Assume that the latch time for the interstage buffers in the pipeline is negligible.

Instruction	t_{EX} in nanoseconds	M
LOAD	1.8	15
IMUL	1.5	10
IDIV	2.5	5
FADD	1.7	10
FSUB	1.7	5
FMUL	2.8	15
FDIV	3.2	5
All other instructions	Less than 1.0	35

When program P is executed, the number of clock cycles for which the pipeline is stalled due to structural hazards in the EX stage is _____. (*answer in integer*)

Q.61	Consider the recursive functions represented by the following code segment:
	
	`int bar(int n){`
	`  if (n == 1) return 0;`
	`  else return 1 + bar(n/2);`
	`}`
	
	`int foo(int n){`
	`  if (n == 1) return 1;`
	`  else return 1 + foo(bar(n));`
	`}`
	
	The smallest positive integer `n` for which `foo(n)` returns 5 is _____. (*answer in integer*)
	
	Note: Ignore syntax errors (if any) in the function.

Q.62	The following sequence corresponds to the preorder traversal of a binary search tree T: 50, 25, 13, 40, 30, 47, 75, 60, 70, 80, 77 The position of the element 60 in the postorder traversal of T is _____. (*answer in integer*) *Note:* The position begins with 1.

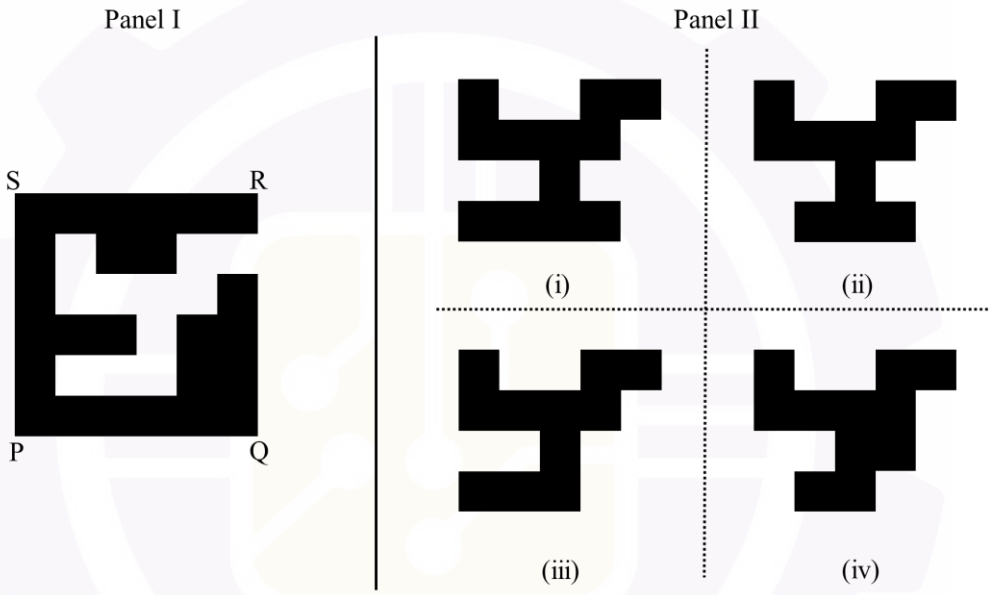
Q.63	Consider the following program snippet. Assume that the program compiles and runs successfully. Further, assume that the `fork()` system call is always successful in creating a process.
	
	`int main () {`
	`  int i;`
	`  for (i = 0; i < 3; i++){`
	`  if (fork() == 0){`
	`  continue;`
	`  }`
	`  break;`
	`  }`
	`  printf("Hello!");`
	`  return 0;`
	`}`
	
	The total number of times that the `printf` statement gets executed is _____. (*answer in integer*)

Q.64	Consider a CPU that has to execute two types of processes. The first type, Actuators (A), requires a CPU burst of 6 seconds. The second type, Controllers (C), requires a CPU burst of 8 seconds. A new process of type A arrives at time t = 10, 20, 30, 40, and 50 (in seconds). Similarly, a new process of type C arrives at time t = 11, 22, 33, 44, and 55 (in seconds). The CPU scheduling policy is First Come First Serve (FCFS). The first process of type A starts running at $t = 10$ seconds. The average waiting time (in seconds) for the 10 processes is _____. (*rounded off to one decimal place*)
Q.65	Consider a relational database schema with a relation $R(A, B, C, D)$. If $\{A, B\}$ and $\{A, C\}$ are the only two candidate keys of the relation R, then the number of superkeys of relation R is _____. (*answer in integer*)

General Aptitude (GA)

Q.1 – Q.5 Carry ONE mark Each

Q.1	Expedite, Hasten, Hurry, _____ Fill the blank by choosing a word with a meaning similar to that of the words given above.
(A)	Accelerate
(B)	Retard
(C)	Provide
(D)	Disable

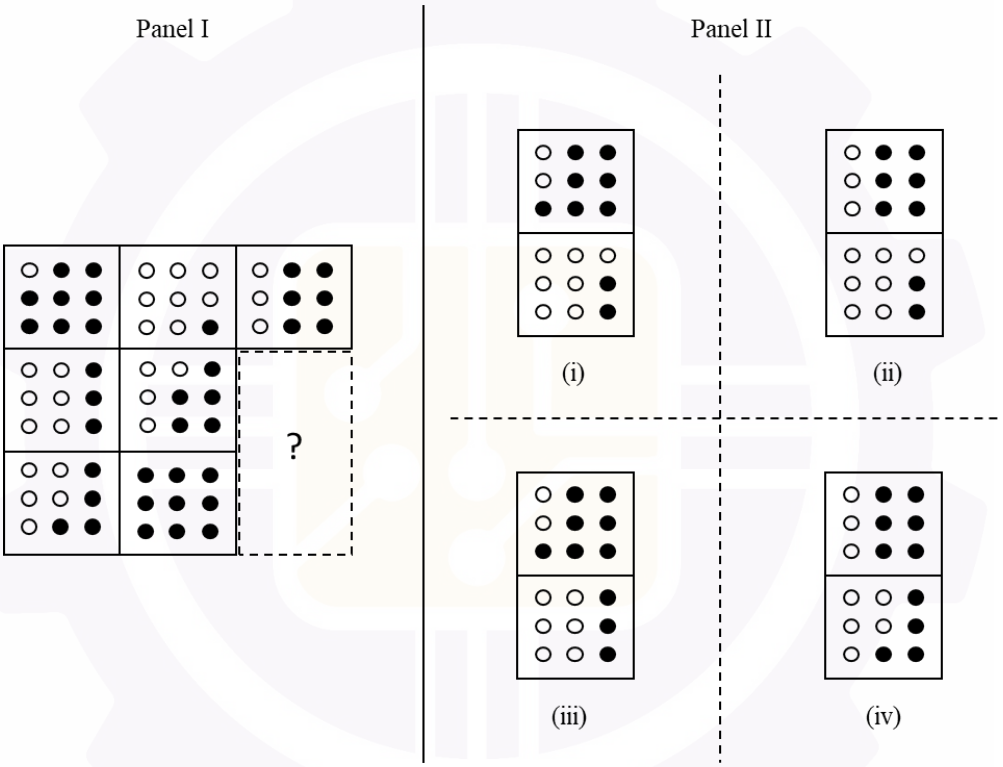
Q.2	A black square PQRS has been cut into two parts. One part of it is shown in Panel I. Which one of the shapes in Panel II is the other part? 
(A)	(i)
(B)	(ii)
(C)	(iii)
(D)	(iv)

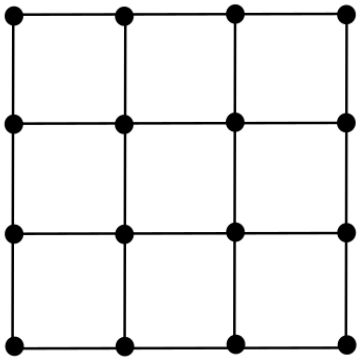
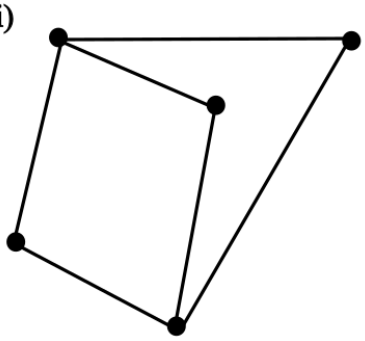
Q.3	A day can only be cloudy or sunny. The probability of a day being cloudy is 0.5, independent of the condition on other days. What is the probability that in any given four days, there will be three cloudy days and one sunny day?
(A)	1/4
(B)	3/4
(C)	2/3
(D)	3/8

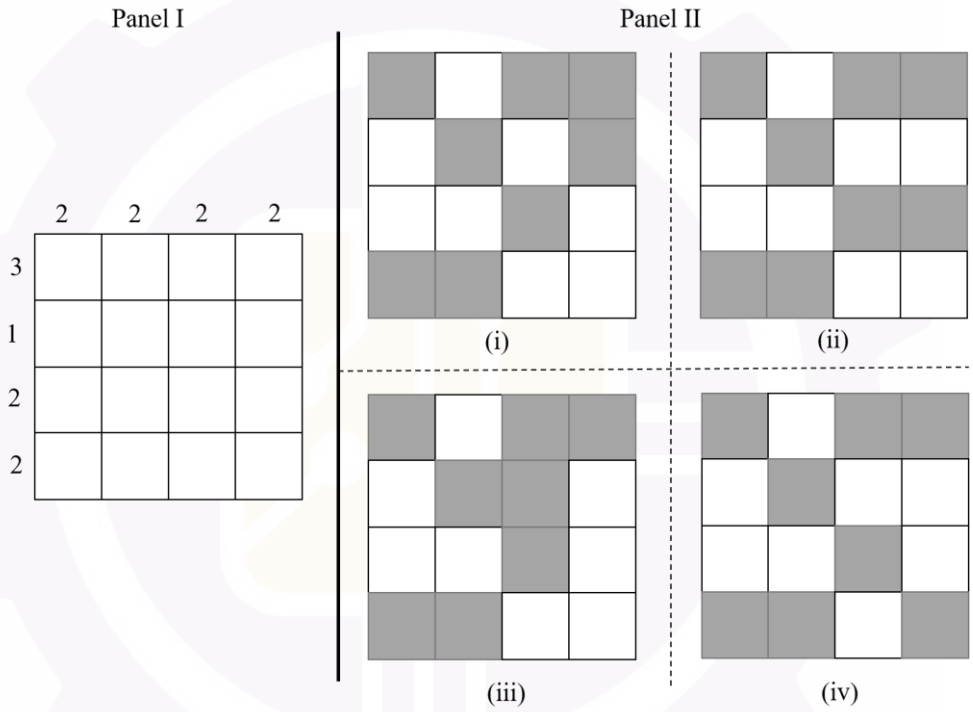
Q. 4	The values of Stock A and Stock B on a particular day are Rs. 50 and Rs. 80, respectively. An investor invests Rs. 100 in Stock A and Rs. 80 in Stock B. He sells all the stocks the next day when the value of Stock A is Rs. 55 and Stock B is Rs. 70. The profit made by the investor is Rs. _____
(A)	0
(B)	5
(C)	10
(D)	20
Q.5	‘When it is raining, peacocks dance.’ Based only on this sentence, which one of the following options is necessarily true?
(A)	Peacocks dance only when it is raining.
(B)	When peacocks dance, it is raining.
(C)	When peacocks are not dancing, it is not raining.
(D)	When it is not raining, peacocks do not dance.

Q.6 – Q.10 Carry TWO marks Each

Q.6	Water : P :: Food : Q Choose the P and Q combination from the options below to form a meaningful analogy.
(A)	P = Thirst; Q = Hunger
(B)	P = Drink; Q = Hunger
(C)	P = Thirst; Q = Satiated
(D)	P = Wet; Q = Critic

Q.7	Two tiles are missing in Panel I. Which one of the options in Panel II is the appropriate choice for the missing tiles? 
(A)	(i)
(B)	(ii)
(C)	(iii)
(D)	(iv)

Q.8	Figures (i) and (ii) represent intercity highway systems. The black dots represent cities and the line segments between them represent intercity highways. A salesperson needs to make a trip. She needs to start from a city, visit each of the remaining cities exactly once, and finally return to the same city from which she started. Which one of the following options is then true? (i)  (ii) 
(A)	Such a trip is possible for (i), but not for (ii).
(B)	Such a trip is possible for (ii), but not for (i).
(C)	Such a trip is possible for both (i) and (ii).
(D)	Such a trip is possible neither for (i) nor for (ii).

Q. 9	The figure in Panel I below is a grid of cells with four rows and four columns. The numbers on the top and on the left represent the number of cells that are to be shaded in that column and row, respectively. Which one of the options shown in Panel II below represents the grid shaded correctly? 
(A)	(i)
(B)	(ii)
(C)	(iii)
(D)	(iv)

Q.10	An unbiased six-faced dice whose faces are marked with numbers 1, 2, 3, 4, 5, and 6 is rolled twice in succession and the number on the top face is recorded each time. The probability that the sum of the two recorded numbers is a prime number is _____
(A)	$\frac{3}{36}$
(B)	$\frac{13}{36}$
(C)	$\frac{15}{36}$
(D)	$\frac{19}{36}$

Q.11 – Q.35 Carry ONE mark Each

Q.11	For two different persons x and y, the predicate $M(x, y)$ denotes that x knows y. Consider the following statement. *There is a person who does not know anyone else, but that person is known by everyone else.* Which one of the following expressions represents the above statement?
(A)	$(\exists y)(\forall x)\ ((x \neq y) \rightarrow (M(x, y) \wedge \neg M(y, x)))$
(B)	$(\forall y)(\exists x)\ ((x \neq y) \rightarrow (M(x, y) \wedge \neg M(y, x)))$
(C)	$(\exists y)(\exists x)\ ((x \neq y) \rightarrow (M(x, y) \wedge \neg M(y, x)))$
(D)	$(\forall y)(\forall x)\ ((x \neq y) \rightarrow (M(x, y) \wedge \neg M(y, x)))$

Q.12 The set **T** represents various traversals over binary tree. The set **S** represents the order of visiting nodes during a traversal.

T	**S**
I: Inorder	L: left subtree, node, right subtree
II: Preorder	M: node, left subtree, right subtree
III: Postorder	N: left subtree, right subtree, node

Which one of the following is the correct match from **T** to **S** ?

(A) I – L, II – M, III – N

(B) I – M, II – L, III – N

(C) I – N, II – M, III – L

(D) I – L, II – N, III – M

Q.13	Which one of the following statements is equivalent to the following assertion? Turing machine M decides the language $L \subseteq \{0,1\}^*$
(A)	Turing machine M halts on all input strings in $\{0,1\}^*$
(B)	Turing machine M accepts all input strings in L
(C)	Turing machine M rejects all input strings in $\{0,1\}^* - L$
(D)	Turing machine M accepts all input strings in L and rejects all input strings in $\{0,1\}^* - L$
Q.14	The probability density function $f(x)$ of a random variable X which takes real values is $$f(x) = \frac{1}{3\sqrt{2\pi}} \exp(-\frac{x^2}{18}), \quad x \in (-\infty, +\infty)$$ Which one of the following statements is correct about the random variable X ?
(A)	X is an exponential random variable
(B)	X is a normal random variable
(C)	X is a Poisson random variable
(D)	X is a uniform random variable

Q.15 In the context of DBMS, consider the two sets **T** and **S** given below.

T	**S**
I: Logical schema	L: Views
II: Physical schema	M: File organization and indexes
III: External schema	N: Relations

Which one of the following is the correct match from **T** to **S** ?

(A) I – L, II – M, III – N

(B) I – M, II – L, III – N

(C) I – N, II – M, III – L

(D) I – N, II – L, III – M

Q.16	Which one of the following options is **not** a property of Boolean Algebra? *Note*: + is OR operation, . is AND operation, and $'$ is NOT operation
(A)	$a + b = b + a$
(B)	$a \,.\, a' = 1$
(C)	$a + a' = 1$
(D)	$a \,.\, b = b \,.\, a$
Q.17	In C runtime environment, which one of the following is stored in heap?
(A)	A static variable declared inside a function
(B)	An array of integers declared inside a function
(C)	A dynamically allocated array of integers created using `malloc()` function call
(D)	Return address of a function

Q.18	Consider the following two statements about interrupt handling mechanisms in a CPU. S1: In non-vectored interrupt mechanism, it usually takes more time to start the Interrupt Service Routine (ISR) when compared to that in a vectored interrupt mechanism. S2: In daisy-chain interrupt mechanism, the CPU polls all the input devices individually to determine the source of the interrupt. Which one of the following options is correct with respect to S1 and S2 ?
(A)	Both S1 and S2 are true
(B)	Both S1 and S2 are false
(C)	S1 is true and S2 is false
(D)	S1 is false and S2 is true

Q.19 Consider the following three ANSI-C programs, P1, P2, and P3.

P1

```
#include <stdio.h>

int a=5;
int main(){
  int a=7;
  return(0);
}
```

P2

```
#include <stdio.h>

int main(){
  int a=5;
  int a=7;
  return(0);
}
```

P3

```
#include <stdio.h>

int main(){
  int a=5;
  float a=7;
  return(0);
}
```

Which one of the following statements is true?

(A) Only P1 will compile without any error

(B) Only P2 will compile without any error

(C) Only P3 will compile without any error

(D) All three programs P1, P2, and P3 will compile without any error

Q.20	Consider concurrent execution of two transactions $T1$ and $T2$ in a DBMS, both of which access a data object A. For these two transactions to not conflict on A, which one of the following statements must be true?
(A)	Both $T1$ and $T2$ only read A
(B)	$T1$ reads A and $T2$ writes A
(C)	$T1$ writes A and $T2$ reads A
(D)	Both $T1$ and $T2$ write A
Q.21	Consider a file of size 4 million bytes being transferred between two hosts connected via a path consisting of three consecutive links of bandwidth 2 Mbps, 500 kbps, and 1 Mbps, respectively. All processing delays and propagation delays are negligible. Assume that there is no other background traffic over the path and no other additional overhead to transfer the file. Which one of the following is the total time (in seconds) to transfer the file? *Note*: $1M=10^6$, $1k=10^3$
(A)	731
(B)	64
(C)	8
(D)	16

Q.22	Which one of the following protocols may need to broadcast some of its messages?
(A)	SMTP
(B)	FTP
(C)	DHCP
(D)	HTTP
Q.23	Which one of the following CPU scheduling algorithms **cannot** be preemptive?
(A)	Shortest Remaining Time First (SRTF) Scheduling
(B)	First Come First Serve (FCFS) Scheduling
(C)	Round Robin Scheduling
(D)	Priority Scheduling

Q.24	Consider the following functions, where n is a positive integer. $$n^{1/3},\ \log(n),\ \log(n!),\ 2^{\log(n)}$$ Which one of the following options lists the functions in increasing order of asymptotic growth rate? *Note*: Assume the base of log to be 2.
(A)	$\log(n),\ n^{1/3},\ 2^{\log(n)},\ \log(n!)$
(B)	$n^{1/3},\ \log(n),\ \log(n!),\ 2^{\log(n)}$
(C)	$\log(n),\ n^{1/3},\ \log(n!),\ 2^{\log(n)}$
(D)	$2^{\log(n)},\ n^{1/3},\ \log(n),\ \log(n!)$

Q.25	Which of the following can be recurrence relation(s) corresponding to an algorithm with time complexity $\Theta(n)$?
(A)	$T(n) = T(n-1) + 1, \quad T(1) = 1$
(B)	$T(n) = 2T\left(\frac{n}{2}\right) + 1, \quad T(1) = 1$
(C)	$T(n) = 2T\left(\frac{n}{2}\right) + n, \quad T(1) = 1$
(D)	$T(n) = T(n-1) + n, \quad T(1) = 1$
Q.26	Let R be a binary relation on the set $\{1, 2, \ldots, 10\}$, where $(x, y) \in R$ if the product of x and y is square of an integer. Which of the following properties is/are satisfied by R?
(A)	Reflexive
(B)	Symmetric
(C)	Transitive
(D)	Antisymmetric

Q.27	For a real number a, let $I(a) = \int_{-1}^{1}(3x^2 - ax + 1)\,dx$. Which of the following statements is/are true?
(A)	The value of $I(a)$ is independent of the value of a
(B)	The value of $I(a)$ can vary with the value of a
(C)	There exists $a \in (-\infty, +\infty)$ such that $I(a)$ is a positive real number
(D)	There exists $a \in (-\infty, +\infty)$ such that $I(a)$ is a negative real number
Q.28	In a system, numbers are represented using 4-bit two's complement form. Consider four numbers $N1 = 1011$, $N2 = 1101$, $N3 = 1010$ and $N4 = 1001$ in the system. Which of the following operations will result in arithmetic overflow?
(A)	$N1 + N2$
(B)	$N2 + N3$
(C)	$N3 - N4$
(D)	$N1 + N4$

Q.29	Which of the following grammars is/are ambiguous?
(A)	$S \rightarrow aSb \mid \epsilon$
(B)	$E \rightarrow E + E \mid E * E \mid id$
(C)	$S \rightarrow aS \mid Sa \mid \epsilon$
(D)	$S \rightarrow aS \mid \epsilon$
Q.30	The keys 5, 28, 19, 15, 26, 33, 12, 17, 10 are inserted into a hash table using the hash function $h(k) = k \ mod \ 9$. The collisions are resolved by chaining. After all the keys are inserted, the length of the longest chain is _____. (*answer in integer*)
Q.31	Consider the system of linear equations given below. $$ax + y = b$$ $$16x + ay = 24$$ Suppose the values of a and b are chosen such that the system of linear equations produce multiple solutions. Then the product of a and b is _____. (*answer in integer*)
Q.32	Consider an array $A = [10, 7, 8, 19, 41, 35, 25, 31]$. Suppose the merge sort algorithm is executed on array A to sort it in increasing order. The merge sort algorithm will carry out a total of 7 merge operations. A merge operation on sorted left array L and sorted right array R is said to be void if the output of the merge operation is the elements of array L followed by the elements of array R. The number of void merge operations among these 7 merge operations is _____. (*answer in integer*)

Q.33	If an IP network uses a subnet mask of 255.255.240.0, the maximum number of IP addresses that can be assigned to network interfaces is _____. (*answer in integer*)
Q.34	The 32-bit IEEE 754 single precision representation of a number is 0xC2710000. The number in decimal representation is _____. (*rounded off to two decimal places*)
Q.35	A lexical analyzer uses the following token definitions • $letter \rightarrow [A-Za-z]$ • $digit \rightarrow [0-9]$ • $id \rightarrow letter\ (letter \mid digit)^*$ • $number \rightarrow digit^+$ • $ws \rightarrow (blank \mid tab \mid newline)^+$ For the string given below, $x1\ \ 23mm\ \ 78\ \ y\ \ 7z\ \ zz5\ \ 14A\ \ 8H\ \ AaYcD$ the number of tokens (excluding ws) that will be produced by the lexical analyzer is _____. (*answer in integer*)

Q.36 – Q.65 Carry TWO marks Each

Q.36	Consider a complete graph K_n with n vertices ($n > 4$). Note that multiple spanning trees can be constructed over K_n. Each of these spanning trees is represented as a set of edges. The Jaccard coefficient between any two sets is defined as the ratio of the size of the intersection of the two sets to the size of the union of the two sets. Which one of the following options gives the lowest possible value for the Jaccard coefficient between any two spanning trees of K_n ?
(A)	$\frac{1}{n}$
(B)	$\frac{1}{2n-3}$
(C)	0
(D)	$\frac{1}{n-1}$

Q.37	Let G be a weighted directed acyclic graph with m edges and n vertices. Given G and a source vertex s in G, which one of the following options gives the worst case time complexity of the fastest algorithm to find the lengths of shortest paths from s to all vertices that are reachable from s in G?
(A)	$\Theta(m+n)$
(B)	$\Theta(m+ n\log(n))$
(C)	$\Theta(nm)$
(D)	$\Theta(n^3)$
Q.38	Consider an array A of integers of size n. The indices of A run from 1 to n. An algorithm is to be designed to check whether A satisfies the condition given below. $\forall i,j \in \{1,\dots,n-1\}$ such that $i>j, (A[i+1]-A[i]) > (A[j+1]-A[j])$ Which one of the following gives the worst case time complexity of the fastest algorithm that can be designed for the problem?
(A)	$\Theta(n)$
(B)	$\Theta(\log(n))$
(C)	$\Theta(n\log(n))$
(D)	$\Theta(n^2)$

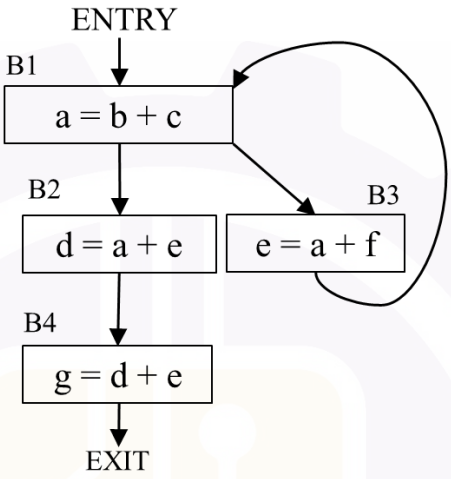
Q.39	Consider a table T, where the elements $T[i][j], 0 \leq i, j \leq n$, represent the cost of the optimal solutions of different subproblems of a problem that is being solved using a dynamic programming algorithm. The recursive formulation to compute the table entries is as follows: $T[0][k] = T[k][0] = 1$ for $k = 0,1,2, \ldots, n$ $T[i][j] = 2T[i-1][j] + 3T[i][j-1]$ for $1 \leq i, j \leq n$ Consider the following two algorithms to compute entries of T. Assume that for both the algorithms, for all $0 \leq i, j \leq n$, $T[i][j]$ has been initialized to 1. Algorithm B_1: For $i = 1, 2, \ldots, n$ For $j = 1, 2, \ldots, n$ $T[i][j] = 2T[i-1][j] + 3T[i][j-1]$ Algorithm B_2: For $s = 2, 3, \ldots, 2n$ For $i = 1, 2, \ldots, n$ For $j = 1, 2, \ldots, n$ If $(i + j == s)$ $T[i][j] = 2T[i-1][j] + 3T[i][j-1]$ Algorithm $B_k, k \in \{1,2\}$ is said to be correct if and only if it calculates the correct values of $T[i][j]$, for all $0 \leq i, j \leq n$, (as per the recursive formulation) at the end of the execution of the algorithm B_k. Which one of the following statements is true?
(A)	Both algorithms B_1 and B_2 are correct
(B)	Algorithm B_1 is correct, but algorithm B_2 is incorrect
(C)	Algorithm B_2 is correct, but algorithm B_1 is incorrect
(D)	Both algorithms B_1 and B_2 are incorrect

Q.40	Consider the following 4-variable Boolean function $$F(A, B, C, D) = \Sigma m(0, 1, 2, 3, 8, 9, 10, 11)$$ Consider A as MSB, D as LSB. Which one of the following options represents the minimal sum of products form for the above function? *Note*: $+$ is OR operation, $.$ is AND operation, $'$ is NOT operation
(A)	$A' + B' + C' + D'$
(B)	B'
(C)	$A'.B' + A.B$
(D)	A'

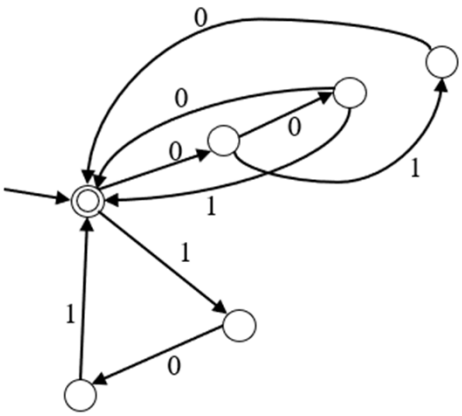
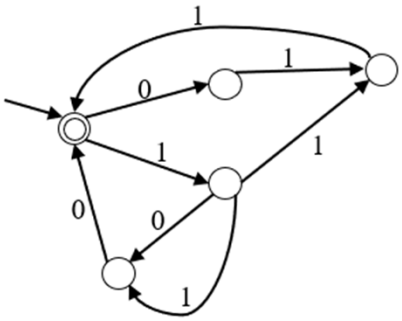
Q.41	Consider the canonical $LR(0)$ parsing of the grammar below using terminals $\{a, b, c\}$ and non-terminals $\{A, B, C, S\}$ with S as the start symbol. $S \rightarrow ACB$ $A \rightarrow aA \mid \epsilon$ $C \rightarrow cC \mid \epsilon$ $B \rightarrow bB \mid b$ Which one of the following options gives the number of shift-reduce conflicts that will occur in the $LR(0)$ ACTION table?
(A)	2
(B)	3
(C)	4
(D)	5

Q.42	In the context of schema normalization in relational DBMS, consider a set F of functional dependencies. The set of all functional dependencies implied by F is called the closure of F. To compute the closure of F, Armstrong's Axioms can be applied. Consider $X, Y,$ and Z as sets of attributes over a relational schema. The three rules of Armstrong's Axioms are described as follows. Reflexivity: If $Y \subseteq X$, then $X \rightarrow Y$ Augmentation: If $X \rightarrow Y$, then $XZ \rightarrow YZ$ for any Z Transitivity: If $X \rightarrow Y$ and $Y \rightarrow Z$, then $X \rightarrow Z$ The additional rule of Union is defined as follows. Union: If $X \rightarrow Y$ and $X \rightarrow Z$, then $X \rightarrow YZ$ It can be proved that the additional rule of Union is also implied by the three rules of Armstrong's Axioms. Listed below are four combinations of these three rules. Which one of these combinations is both necessary and sufficient for the proof ?
(A)	Reflexivity, Augmentation, and Transitivity
(B)	Reflexivity and Augmentation
(C)	Transitivity
(D)	Augmentation and Transitivity

Q.43	Consider the transmission of data bits 110001011 over a link that uses Cyclic Redundancy Check (CRC) code for error detection. If the generator bit pattern is given to be 1001, which one of the following options shows the remainder bit pattern appended to the data bits before transmission?
(A)	011
(B)	101
(C)	000
(D)	100
Q.44	Consider a processor that has 16 general purpose registers and it uses 2-byte instruction format for all its instructions. Variable-sized opcodes are permitted. There are three different types of instructions; M-type, R-type, and C-type. Each M-type instruction has 2 register operands and a 6-bit immediate operand. Each R-type instruction has 3 register operands. Each C-type instruction has a register operand and a 6-bit offset value. If there are 2 unique M-type opcodes and 7 unique R-type opcodes, which one of the following options gives the maximum number of unique opcodes possible for C-type instructions?
(A)	8
(B)	4
(C)	64
(D)	16

Q.45	Consider the control flow graph given below.  Which one of the following options is the set of live variables at the exit point of each basic block?
(A)	B1:{a, b, c, e, f}, B2:{d, e}, B3:{b, c, e, f}, B4:∅
(B)	B1:∅, B2:{d, e}, B3:{a, c, f}, B4:∅
(C)	B1:{a, b, c, e, f}, B2:{d, e}, B3:{c, e, f}, B4:∅
(D)	B1:∅, B2:{d, e, f}, B3:{a, b, c, e, f}, B4:∅

Q.46	An index in a DBMS is said to be dense if an index entry appears for every search-key value in the indexed file. Otherwise it is called a sparse index. Consider the following two statements. S1: A hash index must be a dense index S2: A B^+ tree index can be a sparse index Which one of the following options is correct?
(A)	Both S1 and S2 are true
(B)	Both S1 and S2 are false
(C)	S1 is true and S2 is false
(D)	S1 is false and S2 is true

Q.47	Consider the following two finite automata D_1 and D_2.  D_1  D_2 Which of the following statements is/are true?
(A)	$L(D_1) = L(D_2)$
(B)	$L(D_1)$ is a proper subset of $L(D_2)$
(C)	$L(D_1) \cap L(D_2) = \{\epsilon\}$
(D)	$\left(L(D_1) \cup L(D_2)\right)^*$ consists of all strings in $\{0,1\}^*$ whose length is divisible by 3

Q.48	Let $\Sigma = \{a, b, c, d\}$ and let $L = \{a^i b^j c^k d^\ell \mid i, j, k, \ell \geq 0\}$. Which of the following constraints ensure(s) that the language L is context-free?
(A)	$i + k = j + \ell$
(B)	$i = k$ and $j = \ell$
(C)	$i = \ell$ and $j = k$
(D)	$i + j = k + \ell$

Q.49	Consider a binary search tree (BST) with n leaf nodes ($n > 0$). Given any node V, the key present in the node is denoted as $Val(V)$. All the keys present in the given BST are distinct. The keys belong to the set of real numbers. For a node V, let $Suc(V)$ denote the node that is its inorder successor. If a node V does not have an inorder successor, then $Suc(V)$ is $NULL$. As there are no duplicates, if $Suc(V)$ is not $NULL$, then $Val(V) < Val(Suc(V))$. Corresponding to every leaf node L_i that has a non-NULL $Suc(L_i)$, a new key k_i with the following property is to be inserted into the BST. $$Val(L_i) < k_i < Val(Suc(L_i))$$ Let K represent the list of all such new keys to be inserted into the BST. Which of the following statements is/are true?
(A)	K cannot have any duplicates
(B)	K will have at least one element
(C)	After inserting all keys from K, the height of the BST can increase at most by one
(D)	Number of nodes in the BST will double after inserting all keys from K

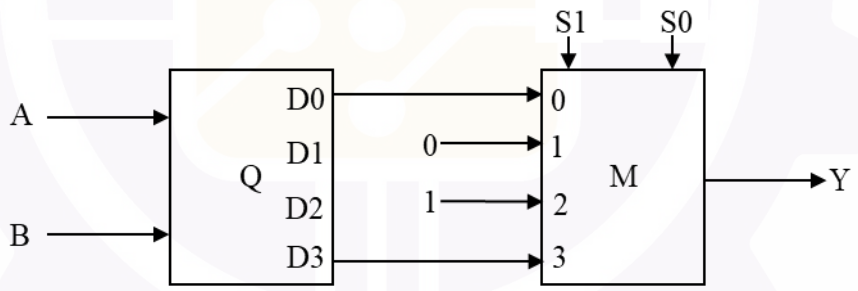
Q.50	Consider a stack S and a queue Q. Both of them are initially empty and have the capacity to store ten elements each. The elements $1, 2, 3, 4,$ and 5 arrive one by one, in that order. When an element arrives, it is assigned either to S (pushed on S) or to Q (enqueued to Q). Once all the five elements are stored, the output is generated in two steps. First, stack S is emptied by popping all elements. Then queue Q is emptied by dequeueing all elements. The output obtained by following this process is 4 3 1 2 5 . Given the output, the objective is to predict whether an element was assigned to S or Q. Which of the following options is/are possible valid assignment(s) of the elements? *Note:* In the options, the notation xS denotes that element x was assigned to S and yQ denotes that element y was assigned to Q.
(A)	$1S, 2Q, 3S, 4S, 5Q$
(B)	$1Q, 2Q, 3S, 4S, 5Q$
(C)	$1Q, 2Q, 3Q, 4S, 5S$
(D)	$1S, 2S, 3S, 4Q, 5Q$

Q.51	Consider three processes P1, P2, and P3 running identical code, as shown in the pseudocode below. A and B are two binary semaphores initialized to 1 and 0, respectively. X is a shared variable initialized to 0. Each line in the pseudocode is executed atomically.
	
	Pseudocode of P1, P2, and P3
	
	`Wait(A);`
	`Print(*);`
	`X = X+1;`
	`If (X == 2)`
	`{`
	`Print($);`
	`Signal(B);`
	`}`
	`Signal(A);`
	`Wait(B);`
	`Print(#);`
	`Signal(B);`
	
	Assume that any of the three processes can start to execute first and context switching can happen between these processes at any arbitrary time and in any arbitrary order.
	
	Which of the following patterns is/are possible to be generated as an outcome of the execution of these three processes?

(A)	**$*###
(B)	**$#*##
(C)	**$##*#
(D)	***$###

Q.52	Consider a system with a processor and a 4 KB direct mapped cache with block size of 16 bytes. The system has a 16 MB physical memory. Four words P, Q, R, and S are accessed by the processor in the same order 10 times. That is, there are a total of 40 memory references in the sequence P, Q, R, S, P, Q, R, S,… Assume that the cache memory is initially empty. The physical addresses of the words are given below (1 word =1 byte). P: 0x845B32, Q: 0x845B26, R: 0x845B36, S: 0x846B32 Which of the following statements is/are true? *Note:* $1K=2^{10}$ and $1M=2^{20}$
(A)	Every access to P results in a cache miss
(B)	Every access to R results in a cache hit
(C)	Every access to Q results in a cache miss
(D)	Except the first access to S, all subsequent accesses to S result in cache hits
Q.53	To keep track of free blocks in a file system, one of the two approaches is generally used – using bitmaps (bit vectors) or using linked lists. Consider that the linked list approach is used to keep track of free blocks in a file system. Assume that the disk size is 16 GB, block size is 2 KB, and block numbers used are 32-bit long. A single pointer of size 4 bytes is used in each block of the list to point to the next block of the list. The number of blocks required to hold the free disk block numbers is _____. (*answer in integer*) *Note*: $1K=2^{10}$ and $1G=2^{30}$

Q.54	A system has a Translation Lookaside Buffer (TLB) that has a reach of 1 MB. TLB reach is defined as the total amount of physical memory that can be accessed through the TLB entries. The paging system uses pages of size 4 KB. The virtual address space is 64 GB and physical address space is 1 GB. If each TLB entry stores a 4-bit process id, page number, frame number, and a 2-bit control field, then the size of the TLB (in bytes) is _____. (*answer in integer*) *Note*: 1K=2^{10}, 1M=2^{20}, 1G=2^{30}
Q.55	Consider contiguous allocation of physical memory to processes using variable partitioning scheme. Suppose there are 8 holes in the memory of sizes 20 KB, 4 KB, 25 KB, 18 KB, 7 KB, 9 KB, 15 KB, and 12 KB. Assume that no two holes are adjacent. Two processes P1 of size 16 KB and P2 of size 9 KB arrive in that order, and they are allocated memory using the best-fit technique. After allocating space to P1 and P2, the number of holes of size less than 8 KB is _____. (*answer in integer*) *Note:* 1K=2^{10}
Q.56	Consider a system with 1 MB physical memory and a word length of 1 byte. The system uses a direct mapped cache, with block numbers starting from 0. The word with physical address 0xA2C28 is mapped to the cache block number 176_{10}. The maximum possible size of the cache (in KB) for this configuration is _____. (*answer in integer*) *Note:* 1K=2^{10} and 1M=2^{20}
Q.57	A non-pipelined instruction execution unit that operates at 1.6 GHz clock takes an average of 5 clock cycles to complete the execution of an instruction. To improve the performance, the system was pipelined with a goal of achieving an average throughput of one instruction per clock cycle. However, it could operate only at 1.2 GHz due to pipeline overheads. While executing a program in the pipelined design, 30% of instructions encountered a stall of 2 cycles due to pipeline hazards. The speed-up obtained by the pipelined design over the non-pipelined one for this program is _____. (*rounded off to two decimal places*) *Note*: 1G=10^9

Q.58	Consider a new TCP connection between a sender and a receiver. The receiver advertised window is constant at 48 KB, the maximum segment size (MSS) is 2 KB, and the slow start threshold for TCP congestion control is 16 KB. Assume that there are no timeouts or duplicate acknowledgements. The number of rounds of transmission required for the congestion control algorithm of the TCP connection to reach the congestion avoidance phase is _____. (*answer in integer*) *Note*: $1K=2^{10}$
Q.59	Consider the digital circuit shown below with two input lines A and B, two select lines S0 and S1, and an output line Y. The blocks Q and M represent active high 2:4 decoder and 4-to-1 multiplexer, respectively. Out of 16 possible input combinations, the number of combinations that produce Y=1 is _____. (*answer in integer*) *Note:* One input combination is an instance of [A B S1 S0]. 

Q.60	Consider the following ANSI-C program.
	
	`#include <stdio.h>`
	
	`int main(){`
	`  int *ptr, a, b, c;`
	`  a=5; b=11; c=20;`
	`  ptr=&a; *ptr=c; ptr=&c;`
	`  a=*(&b); c=*ptr-a;`
	`  printf("%d",c);`
	`  return(0);`
	`}`
	
	The output of this program is _____. (*answer in integer*)
	
	Note: Assume that the program compiles and runs successfully.

Q.61	Consider the following ANSI-C function.
	`int func(int start, int end){`
	`  int length=end+1-start;`
	`  if((length<1)||(start<0)||(end<0)){ return(0); }`
	`  if(length%3==0){`
	`  return(func(start+1, end));`
	`  } else if(length%3==1){`
	`  return(1+func(start, end-1));`
	`  } else {`
	`  return(func(start+2, end));`
	`  }`
	`}`
	The maximum possible value that can be returned from this function is _____. (*answer in integer*)
	Note: Ignore syntax errors (if any) in the function.

Q.62	The determinant of a 4×4 matrix A is 3. The value of the determinant of $2A$ is _____. (*answer in integer*)
Q.63	Suppose an unbiased coin is tossed 6 times. Each coin toss is independent of all previous coin tosses. Let E_1 be the event that among the second, fourth, and sixth coin tosses, there are at least two heads. Let E_2 be the event that among the first, second, third, and fifth coin tosses, there are equal number of heads and tails. The conditional probability $P(E_1 \mid E_2)$ is equal to _____. (*rounded off to one decimal place*)
Q.64	Consider a function $f: (0,1) \rightarrow \{0, 1\}$ defined as follows. For a real number $r \in (0,1)$, $f(r) = 1$ if the second digit after the decimal point in r is one of the four digits 2, 3, 6 and 7. Otherwise, $f(r)$ is equal to 0. The number of points in (0,1) at which f is discontinuous is _____. (*answer in integer*)

Q.65	It is necessary to design a link-layer protocol between two hosts that are directly connected over a lossless link of length 3000 kilometers. Assume that the link bandwidth is 10^8 bits per second and that the propagation delay in the link is 5 nanoseconds per meter. Every transmitted data byte is assigned a unique sequence number. Let N be the minimum number of bits needed for the sequence number field in the protocol header such that i. the sequence numbers do not wrap around before 60 seconds, and ii. the maximum utilization of the link is achieved. The value of N is _____. (*answer in integer*)